

Modification of YOLO with GhostNet to Reduce Parameters and Computing Resources for Detecting Acute Lymphoblastic Leukemia

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Abstract—The Detection of acute lymphoblastic leukemia (ALL) subtypes on multicellular microscopic images is significant for early diagnosis to support the treatment process. Recently, the object detection with a deep learning approach shows good accuracy and fast computation time in the medical field. Therefore, we propose using the You Only Look Once (YOLO) method, namely the Yolov4 and Yolov5 models, to detect the L1, L2, and L3 subtypes. However, both models still have high GFLOPS values and high number of parameters. This paper proposes a modification of Yolov4 and Yolov5 by replacing the standard backbone convolution module with the GhostNet convolution module. The GhostNet module can reduce the GFLOPS value and the number of parameters. Overall, the Yolo backbone modification model has comparable results with the original Yolo model with a slight difference with a value of 1.4% in the Yolov4 backbone modification model and 2.4% in the Yolov5 backbone modification model. The number of parameters and GFLOPS values of the two models modified was reduced by 35% and 40%, respectively.

Keywords— Acute Lymphoblastic Leukemia, Yolov4, Yolov5, GhostNet

I. INTRODUCTION

The process of diagnosing leukemia generally examines microscopic images of multicellular blood. The hematologist can perform a manual analysis or use a computer-aided diagnosis for the examination. The hematologist scrutinizes the morphological characteristics of each white blood cell in order to conclude the type of leukemia present. If the diagnosis results are determined to be ALL, then additional examination is performed regarding the morphology of ALL from the subtypes L1, L2, and L3. Computer-aided diagnosis examination generally requires several stages such as segmentation, separation of touching cells, feature extraction, and classification [1]. The success of the ALL subtype detection process is very dependent on the success of each stage, especially from the stage of separating touching cells.

The object detection approach has recently been widely used in the medical field. The performance of the object detection method is excellent and outperforms conventional methods in terms of accuracy and detection speed. The object detection method does not require many steps and can detect objects in one concise step. An object detection approach can be used to examine the morphology and localization of ALL subtypes. YOLO is a deep learning-based object detection

model that can perform object detection processes in a single learning framework [2], [3]. Research conducted by [4] used Yolov4 to detect white blood cells to determine whether they were ALL or normal.

YOLO has been used in previous studies to detect diseases, especially cancer. Prostate cancer is detected by analyzing the biopsy images for localization and classification of important areas using YOLOv3 [5]. The results of this study can produce detection accuracy values of 97% and 89% in two different datasets. Another study compared Yolov1, Yolov2, and Yolov3 to detect melanoma using measurements of the best mAP values [6]. The comparison results show that Yolov2's best mAP value is 83%, compared to Yolov1 has 37%, and Yolov3 has 77%. The development of the Yolo model into Yolov4 and Yolov5 was used to detect breast tumors in mammogram and ultrasound datasets. Both models outperformed the Yolov3 and Faster R-CNN models [7], [8]. The YOLO model is a popular research topic because it has the advantage of generalizing spatial information on images with direct detection. Nevertheless, Yolo necessitates a relatively sizeable computational value and the number of parameters, which impacts the computational time required for object detection and the model's effectiveness.

Yolo models can be made more efficient by using GhostNet to lighten and reduce the number of parameters. By altering the configuration of a different convolution module on CNN, Han et al. introduced GhosNet in 2021 [9]. GhostNet has the benefit of reducing model performance weight while maintaining high accuracy. The GhostNet model is used in medical research to categorize cervical cell cancer, outperforming Densenet-121 and ShufflenetV2 [8].

This study proposes a GhostNet-based modification of the Yolov4 and Yolov5 models on the backbone to reduce the number of parameters and GFLOPS values, as well as to accelerate the object detection time. Using the GhostNet convolution module, the modification process modifies the standard convolution of the Yolov4 and Yolov5 models. We compared the original Yolov4 and Yolov5 models with our modified model to demonstrate the robustness of the model's performance in detecting ALL subtypes on multi-cell blood images.

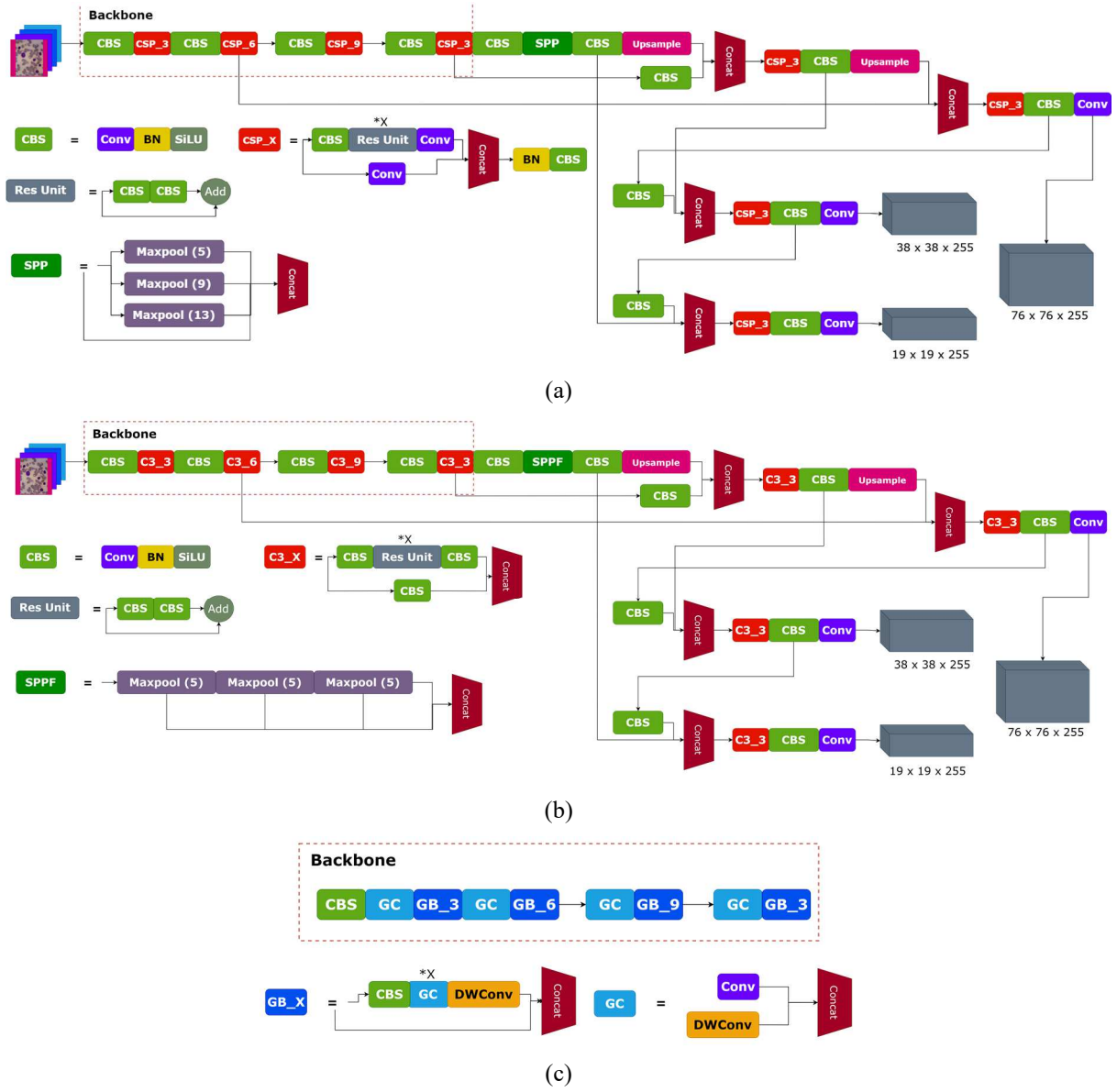


Fig. 1. (a) Yolov4 architecture; (b) Yolov5 architecture; (c) Backbone modification with GhostNet

The remaining papers are organized as follows; Part II describes the research methodology of the proposed methods, datasets and evaluation methods. Section III describes the results and discussion by explaining the experimental findings in visual form and the results of the evaluation metrics. Part IV describes the conclusions of the research results and possible future work.

II. RESEARCH METHODOLOGY

A. Proposed Method

The main backbone of Yolov4 and Yolov5 is CSP (cross-stage partial), which consists of CBS (Convolution, batch normalization, SiLU), residual connection, and cross-stage connection with skip connection containing convolution. The CSP results are combined, and it is recommended to perform batch normalization of the acquired features and end with the CBS process. The architectural details of Yolov4 are shown in Fig. 1(a). In Yolov5, CSP is also used, but there are some modifications. The modification process is done by adding 3 CBS to the CSP, called C3. The results of C3 can contain

more feature maps resulting from convolution because more convolution processes are carried out. The architectural details of Yolov5 are shown in Fig. 1(b). Prior to feature map scaling, CSP and C3 modules were compiled and deployed multiple times on the backbone. The feature map scaling on Yolov4 uses Spatial Pyramid Pooling (SPP) to scale the feature map into several sizes with different pooling sizes. The max-pooling sizes used are 5x5, 9x9, and 13x13. In Yolov5, the feature scaling process is a modified version of SPP called Spatial Pyramid Pooling Fast (SPPF). SPPF consists of three max-poolings with a kernel size of 5x5 sequentially, with each max-pooling result combined at the end of the SPPF.

We modified the Yolov4 and Yolov5 backbones in this study using the GhostNet convolution module. Standard convolution modules are often composed of many layers and neurons, which result in enormous computational costs and redundant intermediate feature maps. GhostNet's convolution module reduces computational costs by decreasing the convolution filter used to generate map features. The process of generating map features is influenced by the number of

input channels, the height and width of the input data, convolution filters and kernel size. Details of the map feature calculation process are shown in Eq. 1. During the convolution process, the required number of floating point operations per second (FLOPs) can be calculated using Eq. 2. On Eq. 2, n is the number of classes, h' is the height of the output dimension, w' is the width of the output dimension, c is the number of input channels, and k is the kernel. FLOPs values can reach hundreds of thousands or even millions, depending on the number of filters and channels from the convolution layer, which are generally very large.

$$Y = X * f + b \quad (1)$$

$$FLOPs = n * h' * w' * c * k \quad (2)$$

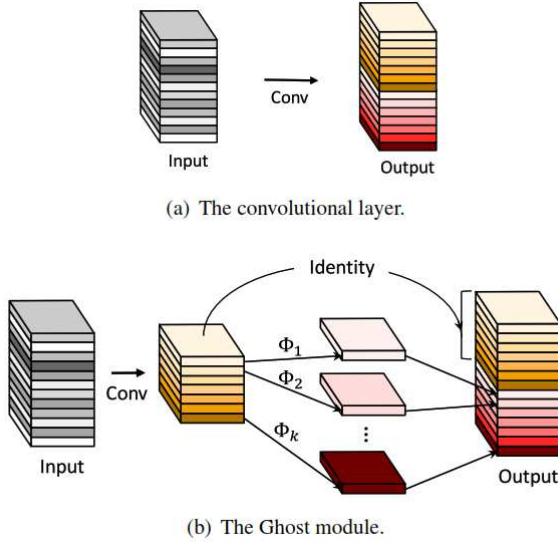


Fig. 2. The difference between standard convolution and ghostnet convolution [9]

The result of Eq. 1, mainly from the filter and bias, is affected by the dimensions of the input and output features of the map. The convolution process, as shown in Fig. 2(a), produces multiple redundant feature map outputs which contain the same information. The result of the redundant map feature can increase the FLOPs value and the number of parameters of the convolution module. The Ghostnet convolution module, which replaces the standard convolution module, can reduce the value of FLOPs and the number of parameters by optimizing filters and biases. The output of the feature map from the ghostnet convolution module is called the intrinsic feature map. The number of Intrinsic map features is less than the standard map features obtained using ordinary convolution filters. Calculating the intrinsic map feature can use Eq. 3. on Eq. 3, f' is the number of filters used with the number of intrinsic map features resulting from the ordinary convolution filter. The results of the intrinsic features are carried out with linear operations to produce more map features. Then the intrinsic feature map is carried out by identity mapping, which ends with the stacking process of each map feature, as shown in Fig. 2(b).

$$Y = X * f' \quad (3)$$

We modified the CSP bottleneck on our backbone with Ghost Bottleneck, which consists of CBS, GhostNet convolution modules, and Depthwise convolutions. Depthwise convolution is a convolution process that is done

by dividing the channel into individual channels and then applying a convolution filter to each individual channel. The results of the depthwise convolution are less computationally valuable than the standard convolution process. The Ghost Bottleneck results in the merging process by passing connections from the initial feature map. The results of the backbone modification can reduce the number of parameters and GFLOPs values so that it can speed up the computation time of object detection. The architectural details of the backbone modification are shown in Fig. 1(c).

B. Dataset

This study utilized a multi-cell blood microscopic image dataset containing information on the L1, L2, and L3 subtypes sourced from the paper of Fatonah et al. [10], which focuses on segmenting and classifying using conventional machine learning models. According to the literature, the dataset's source is a hospital with detailed images taken at 1000x magnification using the same staining technique. 128 L1, 63 L2 images, and 110 L3 images comprise 301 multi-cell subtypes of ALL image data. The annotation uses the YOLO format $[x, y, w, h]$, where x and y represent the centre point of the bounding box, and w and h represent the height and width of the bounding box, respectively.

The preprocessing stage in this study begins with manual labelling using roboflow based on data obtained from paper [10]. The preprocessing stage also includes resizing from 591 x 443 pixels to 416 x 416 pixels. We chose the 416 x 416 size because it is the standard often used on Yolo models. The manual labelling process conforms to the Yolo annotation format, which is grouped into L1, L2, and L3.

C. Evaluation Metric

Experiment results from the evaluation phase of the model's performance in detecting and classifying ALL subtypes on multi-cell blood microscopy images were used to determine the model's performance. This study employs precision, recall, F1, mean average precision (mAP), and computation time as evaluation metrics. The objective of the precision measurement procedure is to compare the True Positive prediction to the overall positive predicted model results. Recall provides information for comparing True Positive predictions to the detailed data that predicted True Positive. F1 measures the proportion of the average comparison of recall and precision values weighted by Eq. 4 to generate precision values, Eq. 5 to generate recall values, and Eq. 6 to generate F1.

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive} \quad (4)$$

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative} \quad (5)$$

$$F1 = \frac{2 \times (Recall \times Precision)}{Recall + Precision} \quad (6)$$

Combining recall and precision, average precision is a measurement result [11]. In other words, it evaluates the accuracy of all documents' or output data's predictions concerning the actual data. True Positive indicates that an object with the correct classification has been detected and the IoU value is more significant than specified, such as $IoU \geq 0.5$ or 50%. False Positive refers to the detection of an object that does not conform to the classification and has an IoU below the threshold value. False Negative indicates the number of non-detected objects based on ground truth data.

Eq. 7 shows the Average precision formula. The calculation of precision and recall yields the Average Precision (AP) score, which is the average precision value of the total recall, where r represents the recall value and p represents the precision value. As shown in Eq. 8, the mAP is computed by determining the AP for each class and then averaging the results across multiple classes. In this study, the thresholds used were 0.5 and 0.95.

The evaluation of computing time involves the search for a single bounding box based on four criteria: preprocessing, inference, Non-maximum Suppression (NMS), and total. The preprocessing and inference computation time is the time from the beginning of the image to the beginning of the preprocessing phase and the detection of the ALL subtype. The time required to compute inference is greater than that of other variables because the detection process tries numerous object combinations. The NMS chooses each bounding box with the highest filter confidence and one that overlaps.

$$AP = \sum_{k=0}^{k=n-1} [Recalls(k) - Recalls(k+1)] \quad (7)$$

$* Precisions(k)$

$n = \text{Number of threshold}$

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (8)$$

III. RESULTS AND DISCUSSIONS

This study uses stratified k-fold cross-validation for the data resampling procedure because the amount of data is still small, and the proportion of the number of each class is not balanced. The entire dataset is divided into five folds by considering each fold's proportion of class division. The division process is divided evenly from the three ALL subtype classes so that the model can generalize from each feature in each class. The data resampling scenario process was performed five times by distributing different training data, validation data, and testing data. Details of the distribution of the data resampling scenarios are shown in Fig. 3. The experimental stages in this study were divided into four, namely Yolov4 original, Yolov5 original, Yolov4 modified backbone, and Yolov5 modified backbone. Details of the research and experimental stages are shown in Fig. 4.

Train	Train	Train	Val	Test
Train	Train	Val	Test	Train
Train	Val	Test	Train	Train
Val	Test	Train	Train	Train
Test	Train	Train	Train	Val

Fig. 3. Data Resampling Scenario Details

The original Yolov4 and Yolov5 models used standard block convolution without modification. The CSP and C3 modules on the backbone are used to extract feature maps from images. The experimental scenario on Yolov4 and Yolov5 has been modified to replace standard convolution with the GhostNet convolution module. CSP and C3 were modified and replaced with a combination Ghost Bottleneck module, which includes Ghost convolution and Depthwise convolution. The test data is used to compare the results of

the four scenarios as the best model to generalize the features of all subtype cells. Recall values, precision, F1, mAP, number of parameters, GFLOPS, and object detection time per image are all evaluation metrics. The mAP value was calculated using an IoU threshold of 50% and ranged from 50% to 95%, with an increase of 5%. The selection of a 50% threshold is based on the PASCAL VOC metric, and a 50% to 95% threshold selection is based on the COCO metric.

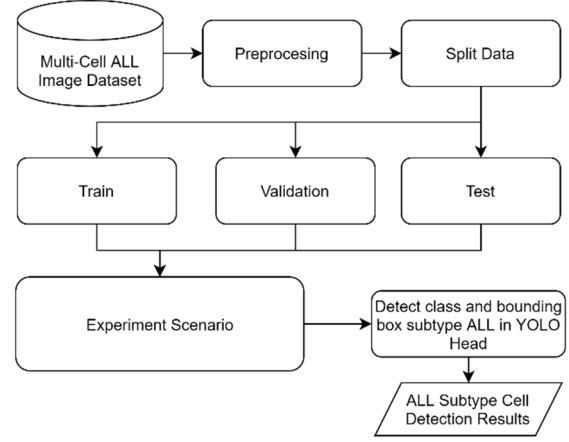


Fig. 4. Flowchart of research experiment and scenario

Fig. 5 shows sample data from the experimental results of four Yolo models in visual form. The original Yolov4 model shows two false positives, L2, which should be L1. The Yolov4 backbone modification model also showed the same two false positives as the original Yolov4 model but with different locations. The original Yolov5 model showed three false positives, L2, which should have been L1. The Yolov5 backbone modification model showed no false positives but redundant results from L1. False positive results from L2 are possible because the morphological structure of the L1 and L2 subtypes is almost identical. To test the performance of the four models, each evaluation metric is compared using the average of the five folds in Table I.

TABLE I. OBJECT DETECTION EXPERIMENT RESULTS

Object	Method	Yolov4 Ori.	Yolov5 Ori.	Yolov4 Mod.	Yolov5 Mod.
ALL	Presisi	89.5%	88.4%	88.9%	85.5%
	Recall	89.5%	88.6%	86.7%	87.0%
	F1	89.5%	88.3%	87.6%	86.1%
	mAP (0.5)	93.2%	92.7%	90.9%	91.5%
	mAP (0.5:0.95)	65.9%	66.2%	65.0%	65.2%
L1	Presisi	89.6%	93.0%	90.3%	87.4%
	Recall	95.0%	93.3%	94.0%	95.7%
	F1	92.1%	92.9%	92.0%	90.9%
	mAP (0.5)	96.3%	96.8%	94.6%	95.5%
	mAP (0.5:0.95)	68.3%	71.3%	67.7%	69.0%
L2	Presisi	87.1%	79.0%	83.9%	79.1%
	Recall	77.6%	79.5%	70.3%	69.6%
	F1	81.9%	75.4%	73.6%	72.0%
	mAP (0.5)	85.4%	83.6%	80.9%	81.4%
	mAP (0.5:0.95)	62.4%	60.3%	60.5%	63.7%
L3	Presisi	91.7%	93.2%	92.5%	90.1%
	Recall	96.0%	92.9%	95.8%	95.6%
	F1	93.8%	92.9%	94.1%	92.4%

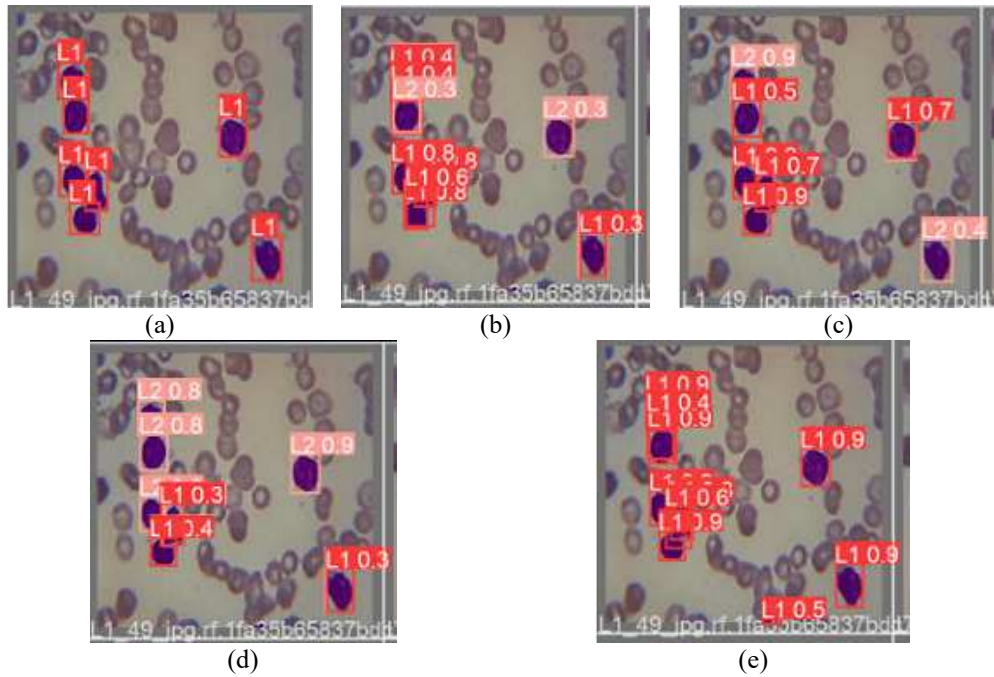


Fig. 5. (a) Ground Truth; (b) Yolov4 original; (c) Yolov4 backbone modification; (d) Yolov5 original; (e) Yolov5 backbone modification

	mAP (0.5)	97.7%	97.9%	97.4%	97.6%
	mAP (0.5:0.95)	66.9%	67.1%	67.0%	67.0%

The experimental results to detect the L1 subtype are superior to the original Yolov5 model compared to the modified backbone model, except for the recall value model, which is 0.7% superior to the modified Yolov5 backbone model. The most significant difference in the evaluation metrics between the original Yolo and Yolo backbone modification models is the mAP(0.5:0.95) metric between the Yolov4 backbone modification and the original Yolov5 models by 3.6%. From the overall average comparison of evaluation metrics between the original Yolo model and the Yolo backbone modification model, there is a difference of 1.2% from the Yolov4 backbone modification model and 0.6% from the Yolov5 backbone modification model.

The results of object detection of the L2 subtype have varying results with three metrics superior to the original Yolov4 model, including precision, F1, and mAP(0.5), one in the original Yolov5 model including recall, and one in the Yolov5 backbone modification model including mAP(0.5:0.95). The most significant difference from the evaluation metrics is shown in the F1 metric between the Yolov5 backbone modification model and the two original Yolov4 and Yolov5 models with a value of 9.8%. The overall average comparison of evaluation metrics from the original model and the backbone modification model shows the superiority of the Yolov4 backbone modification model compared to the original Yolov5 model with a value of 3.4%.

The experimental results of the L3 subtype detection also have mixed results with three evaluation metrics superior to the original Yolov5 model, including precision, mAP(0.5), and mAP(0.5:0.95). The most significant difference between the original Yolo model and the Yolo modification backbone is shown in the recall evaluation metric with a value of 2.9%

which can outperform the original Yolov4 model and the original Yolov5 model.

Overall, the modified Yolo backbone model has comparable results with the original Yolo model with a slight difference with a value of 1.4% for the modified Yolov4 backbone model and 2.4% for the modified Yolov5 backbone model from the values of precision, recall, F1, mAP (0.5) and mAP (0.5:0.95). The recall values from the source paper dataset [10] from L1, L2 and L3, respectively, were 90.63%, 77.78% and 56.36%. The Yolo model can outperform the L1 evaluation by a difference of 2.64% using the original Yolov5 model. The original Yolo model and the modified backbone can outperform the average L2 and L3 evaluations by the biggest differences, 9.32% and 36.84%, respectively. Apart from the evaluation metrics in Table I, we also tested the model's performance in terms of time and computational complexity based on the GFLOPS value, the number of parameters, and object detection time per the image shown in Table II.

TABLE II. COMPARISON OF GFLOPS AND NUMBER OF PARAMETERS

Method	Yolov4 Ori.	Yolov5 Ori.	Yolov4 Mod.	Yolov5 Mod.
GFLOPS	16.3	15.8	9.8	9.6
Parameter	7,210,824	7,018,216	4,692,312	4,588,184
Pre-processing (ms)	0.56	0.62	0.64	0.88
Inference (ms)	5.74	4.82	3.94	5.08
NMS (ms)	2.82	3.32	2.44	3.08
Total (ms)	9.12	8.76	7.02	9.04

Both Yolo backbone modification models can reduce the GFLOPS value by about 40% from the original Yolo model so that the Yolo backbone modification model can save computational complexity. The smallest GFLOPS value is shown by the Yolov5 modification backbone model with a

value of 9.6. The Yolo backbone modification model's parameter values can also decrease by 35% from the original Yolo model. The object detection time evaluation metric shows that the Yolov4 backbone modification model can outperform all models with the shortest time of 7.02 ms. Although the Yolov5 backbone modification model is superior to the GFLOPS value and the number of parameters compared to the Yolov4 backbone modification model, the object detection time is higher by a difference of 2.02 ms. So that the Yolov4 modification backbone model can be used as a model for ALL subtype detection, which is lighter, faster, and slightly different from the original Yolo model.

IV. CONCLUSIONS

The results of our experiment with backbone modification using GhostNet on the Yolov4 and Yolov5 models showed that the model becomes lighter than the original by reducing the GFLOPS value and the number of parameters. Standard convolutions are changed to GhostNet convolutions, which combine depthwise and standard convolutions, as part of the modification process. With a value of 1.4% for the Yolov4 backbone modification model and 2.4% for the Yolov5 backbone modification model, the Yolo backbone modification model generally produces comparable results to the original Yolo model. The Yolov4 and Yolov5 models' GFLOPS value and the total number of parameters can drop by about 40% and 35%, respectively. In the future, further research is needed to maximize the performance of the modified backbone model with GhostNet so that it can exceed all aspects of evaluation.

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